04 — Calendar effects

Scope. All findings describe the visible subset of pharmacy revenue (counter cash receipts, about 65%); see NB01 for the recording basis. 25 December 2019 (an institutional credit-settlement day, per NB01) is excluded throughout.

This notebook fits a single joint OLS regression on log revenue and reads off calendar effects (day-of-week, month-phase, Ramadan and Eid al-Adha) while controlling for the four regimes identified in NB02 (24 March, 28 May and 4 December 2020). Standard errors are HAC (Newey–West) with a 21-lag bandwidth: three weekly cycles, matching NB02, a reasonable constant not tuned to the results; sensitivity from 7 to 28 lags is reported in the diagnostics. Effects are reported as exp(β) − 1, as percentages.

The primary family of tests (Friday effect, start-of-month effect, last-five-days effect) is Holm-adjusted for family-wise error at α = 0.05. All three were chosen after the NB01 EDA, so Holm controls the family-wise error within them, but the analysis is exploratory: it confirms patterns visible in the same data under a better-specified model, not independent evidence.

Religious-calendar scope. Ramadan (a month-long religious observance, not a public holiday) and Eid al-Adha (public holiday) are modelled. Public holidays such as New Year and Labour Day are not modelled here; their potential effects sit inside the residuals.

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from scipy import stats

from pharmacy_covid_analysis.bootstrap import holm_adjust

sns.set_theme(style="whitegrid", context="notebook")
plt.rcParams["figure.figsize"] = (11, 4)
plt.rcParams["axes.titleweight"] = "bold"
```

In [2]:
```
df = pd.read_csv("../data/daily_sales_final.csv", parse_dates=["date"])
df = df.sort_values("date").set_index("date").asfreq("D")

# Per NB01, 25 December 2019 is an institutional credit-settlement day, not a
# regular trading day. Exclude it here so it does not distort calendar effects.
df.loc[pd.Timestamp("2019-12-25"), "revenue_scaled"] = np.nan
df = df.dropna(subset=["revenue_scaled"])

df["log_y"] = np.log(df["revenue_scaled"])
print(
```

```
    f"Observed days: {len(df)} from {df.index.min().date()} to {df.inde
x.max().date()}"
)
```

```
Observed days: 469 from 2019-12-15 to 2021-05-10
```

Regime and calendar flags

Regime boundaries match NB02's K=3 fit. Day-of-week uses Tuesday as the reference (a mid-week working day in the Sunday–Thursday working week here). Weekend days are Friday and Saturday; the Friday effect is the `d_Fri` coefficient.

Month-phase splits every month into a start-of-month block (days 1–5), a mid-month baseline (6 to days-in-month minus 5), and a last-five-days block. Ramadan and Eid al-Adha dummies mark the Ramadan month (religious observance, not a public holiday) and the two Eid al-Adha days (public holiday). Eid al-Fitr 2020 falls inside the 19–25 May missing-data week and Eid al-Fitr 2021 falls after the sample ends, so no Eid al-Fitr dummy is fitted (both would be identified by zero observations).

In [3]:
```
# Regime dummies (base = pre-COVID)
df["r_acute"] = ((df.index >= "2020-03-24") & (df.index <= "2020-05-2
7")).astype(int)
df["r_rebound"] = ((df.index >= "2020-05-28") & (df.index <= "2020-12-0
3")).astype(int)
df["r_plateau"] = ((df.index >= "2020-12-04") & (df.index <= "2021-05-1
0")).astype(int)

# Day-of-week dummies (base = Tuesday). Weekend in this working week is
Fri-Sat.
DOW_NAMES = {0: "Mon", 1: "Tue", 2: "Wed", 3: "Thu", 4: "Fri", 5: "Sat",
6: "Sun"}
df["dow"] = df.index.dayofweek.map(DOW_NAMES)
for day in ["Mon", "Wed", "Thu", "Fri", "Sat", "Sun"]:  # Tuesday omitte
d as base
    df[f"d_{day}"] = (df["dow"] == day).astype(int)

# Month-phase dummies (base = mid-month, days 6 to days-in-month minus
5)
df["day_of_month"] = df.index.day
df["days_in_month"] = df.index.days_in_month
df["start_of_month"] = (df["day_of_month"] <= 5).astype(int)
df["last_5_days"] = (df["day_of_month"] >= df["days_in_month"] - 4).asty
pe(int)


# Religious-calendar dummies (Ramadan 2020, 2021; Eid al-Adha 2020 only;
see markdown above)
def _in_range(idx, start, end):
    return (idx >= pd.Timestamp(start)) & (idx <= pd.Timestamp(end))


df["ramadan"] = (
    _in_range(df.index, "2020-04-24", "2020-05-23")
    | _in_range(df.index, "2021-04-13", "2021-05-12")
).astype(int)
df["eid_al_adha"] = _in_range(df.index, "2020-07-31", "2020-08-01").asty
```

```
pe(int)

# Flag counts
counts = {
    "Pre-COVID": int((df.index < "2020-03-24").sum()),
    "Acute crash": int(df["r_acute"].sum()),
    "Partial rebound": int(df["r_rebound"].sum()),
    "Lower plateau": int(df["r_plateau"].sum()),
    "Friday days": int(df["d_Fri"].sum()),
    "Saturday days": int(df["d_Sat"].sum()),
    "Start-of-month": int(df["start_of_month"].sum()),
    "Last 5 days": int(df["last_5_days"].sum()),
    "Ramadan days": int(df["ramadan"].sum()),
    "Eid al-Adha days": int(df["eid_al_adha"].sum()),
}
print("Observation counts by flag:")
for k, v in counts.items():
    print(f"  {k:<20} {v:>4}")
```

```
Observation counts by flag:
  Pre-COVID              92
  Acute crash            58
  Partial rebound       176
  Lower plateau         143
  Friday days            67
  Saturday days          67
  Start-of-month         79
  Last 5 days            71
  Ramadan days           52
  Eid al-Adha days        2
```

Joint calendar model

The specification is

$\log(\text{revenue}) = \alpha + \Sigma\,\text{regime} + \Sigma\,\text{day-of-week} + \Sigma\,\text{month-phase} + \Sigma\,\text{religious-calendar} + \varepsilon,$

fitted by OLS with HAC standard errors (21 lags).

In [4]:

```
REGRESSORS = [
    "r_acute",
    "r_rebound",
    "r_plateau",
    "d_Mon",
    "d_Wed",
    "d_Thu",
    "d_Fri",
    "d_Sat",
    "d_Sun",
    "start_of_month",
    "last_5_days",
    "ramadan",
    "eid_al_adha",
]

y = df["log_y"]
X = sm.add_constant(df[REGRESSORS])
```

```
model = sm.OLS(y, X).fit(cov_type="HAC", cov_kwds={"maxlags": 21})

print(
    f"n = {int(model.nobs)}, R² = {model.rsquared:.3f}, adj R² = {model.
rsquared_adj:.3f}"
)
```

```
n = 469, R² = 0.660, adj R² = 0.650
```

In [5]:

```
def coef_table(model, names):
    rows = []
    ci = model.conf_int()
    for name in names:
        b = model.params[name]
        se = model.bse[name]
        z = model.tvalues[name]
        p = model.pvalues[name]
        lo, hi = ci.loc[name]
        rows.append(
            {
                "coefficient": name,
                "β": b,
                "HAC SE": se,
                "exp(β)−1 (%)": (np.exp(b) - 1) * 100,
                "HAC z": z,
                "p-value": p,
                "95% CI lower (%)": (np.exp(lo) - 1) * 100,
                "95% CI upper (%)": (np.exp(hi) - 1) * 100,
            }
        )
    return pd.DataFrame(rows)


coefs = coef_table(model, REGRESSORS)
with pd.option_context(
    "display.float_format", "{:+.4f}".format, "display.max_colwidth", No
ne
):
    print(coefs.to_string(index=False))
```

```
   coefficient       β  HAC SE  exp(β)−1 (%)    HAC z  p−value  95% CI low
er (%)  95% CI upper (%)
       r_acute −0.4885 +0.0264      −38.6435 −18.5302  +0.0000          −4
1.7330          −35.3901
     r_rebound −0.1302 +0.0301      −12.2083  −4.3238  +0.0000          −1
7.2399           −6.8708
     r_plateau −0.2556 +0.0294      −22.5562  −8.6818  +0.0000          −2
6.8988          −17.9557
         d_Mon −0.0000 +0.0188       −0.0046  −0.0025  +0.9980           −
3.6215           +3.7480
         d_Wed −0.0053 +0.0227       −0.5297  −0.2342  +0.8149           −
4.8545           +3.9917
         d_Thu −0.0003 +0.0203       −0.0316  −0.0156  +0.9876           −
3.9282           +4.0229
         d_Fri −0.2529 +0.0289      −22.3440  −8.7647  +0.0000          −2
6.6136          −17.8261
         d_Sat +0.0297 +0.0169       +3.0182  +1.7563  +0.0790           −
0.3442           +6.4940
         d_Sun +0.0290 +0.0211       +2.9403  +1.3710  +0.1704           −
1.2371           +7.2944
start_of_month −0.0032 +0.0162       −0.3145  −0.1951  +0.8453           −
3.4207           +2.8915
   last_5_days −0.0199 +0.0193       −1.9662  −1.0298  +0.3031           −
5.6022           +1.8097
       ramadan −0.0325 +0.0253       −3.2022  −1.2861  +0.1984           −
7.8860           +1.7197
   eid_al_adha −0.5119 +0.0246      −40.0648 −20.8187  +0.0000          −4
2.8848          −37.1056
```

Regime effects. All three regime dummies are large and precisely estimated (|HAC z| ≥ 4). Relative to pre-COVID: acute crash −38.6%, partial rebound −12.2%, lower plateau −22.6%. These reproduce NB02's regime picture inside a model that also controls for day-of-week, month-phase and the religious calendar.

Day-of-week effects. The Friday coefficient is −22.3% relative to Tuesday (HAC z = −8.8), but this is an average across regimes; the Friday × regime section below tests whether it conceals regime variation. Monday, Wednesday and Thursday are within about half a percent of Tuesday. Saturday and Sunday sit slightly *above* Tuesday at +3.0% and +2.9%, both not distinguishable from zero at conventional levels. Saturday, the second day of the local weekend, shows no dip: the Friday effect is specific to the weekend's first day.

Month-phase effects. Under this specification, start-of-month is −0.3% and last-five-days is −2.0%; both point estimates are small and neither reaches conventional significance. Both are much smaller than the corresponding NB01 descriptive comparisons because NB01's raw comparisons ran without regime levels or religious-calendar controls, so the raw month-phase differences partly reflect those.

Religious-calendar effects. Ramadan is −3.2% (HAC z = −1.29), a small effect not distinguishable from zero, consistent with NB02's joint-model finding. Eid al-Adha 2020 (31 July, 1 August) is estimated at −40.1% relative to a Friday–Saturday weekend that the model already controls for; the coefficient therefore captures what is unusual about those two days *beyond* the weekend effect. Two things could produce that: NB01's records note reduced trading hours on 31 July, and the two

dates fall inside the partial-rebound regime where the Friday effect is deeper than the pooled model allows for. Only two observations identify the coefficient, so it is a very local descriptive figure, not a general Eid estimate, and its standard error, z and interval in the table are not meaningful.

Residual autocorrelation

To see how much of the residual autocorrelation the regime dummies absorb, the model is refitted with a single COVID dummy and the Ljung–Box statistics of the two specifications are compared.

In [6]:
```
# Single-COVID-dummy specification (for comparison only)
df["covid_flag"] = (df.index >= "2020-03-24").astype(int)
REGRESSORS_SIMPLE = [
    "covid_flag",
    "d_Mon",
    "d_Wed",
    "d_Thu",
    "d_Fri",
    "d_Sat",
    "d_Sun",
    "start_of_month",
    "last_5_days",
    "ramadan",
    "eid_al_adha",
]
X_simple = sm.add_constant(df[REGRESSORS_SIMPLE])
model_simple = sm.OLS(y, X_simple).fit(cov_type="HAC", cov_kwds={"maxlags": 21})

lb_rows = []
for lag in [7, 14, 21]:
    lb_s = sm.stats.acorr_ljungbox(model_simple.resid, lags=[lag], return_df=True).iloc[
        0
    ]
    lb_r = sm.stats.acorr_ljungbox(model.resid, lags=[lag], return_df=True).iloc[0]
    lb_rows.append(
        {
            "lag": lag,
            "single-COVID-dummy LB": lb_s["lb_stat"],
            "single-COVID-dummy p": lb_s["lb_pvalue"],
            "regime-dummies LB": lb_r["lb_stat"],
            "regime-dummies p": lb_r["lb_pvalue"],
        }
    )
lb_df = pd.DataFrame(lb_rows)
with pd.option_context("display.float_format", "{:.4g}".format):
    print("Ljung-Box on residuals (H0: no autocorrelation up to lag k):")
    print(lb_df.to_string(index=False))
```

```
Ljung-Box on residuals (H0: no autocorrelation up to lag k):
 lag  single-COVID-dummy LB  single-COVID-dummy p  regime-dummies LB  regi
me-dummies p
   7                  693.2            2.036e-145              100.6
8.095e-19
  14                   1054            3.319e-216              111.7
2.608e-17
  21                   1343            1.618e-271              116.7
2.91e-15
```

Replacing the single COVID dummy with the four-regime structure cuts the Ljung-Box statistic at lag 7 from about 690 to about 100. Residual autocorrelation is still non-zero — which is why HAC standard errors are used on top — but the regime dummies removed the bulk of it.

Primary family: Friday, start-of-month, last-five-days

Three tests, Holm-adjusted for family-wise error at α = 0.05; see the introduction on why the family is exploratory.

In [7]:
```
PRIMARY = ["d_Fri", "start_of_month", "last_5_days"]
primary_p = model.pvalues[PRIMARY].values
primary_rows = []
for name, p_raw, p_holm in zip(PRIMARY, primary_p, holm_adjust(primary_
p)):
    b = model.params[name]
    lo, hi = model.conf_int().loc[name]
    primary_rows.append(
        {
            "test": name,
            "exp(β)−1 (%)": (np.exp(b) - 1) * 100,
            "95% CI (%)": f"[{(np.exp(lo) - 1) * 100:+.2f}%, {(np.exp(h
i) - 1) * 100:+.2f}%]",
            "HAC z": model.tvalues[name],
            "raw p": p_raw,
            "Holm-adjusted p": p_holm,
        }
    )
primary_df = pd.DataFrame(primary_rows)
with pd.option_context(
    "display.float_format", "{:.4f}".format, "display.max_colwidth", Non
e
):
    print(primary_df.to_string(index=False))
```

```
         test  exp(β)−1 (%)         95% CI (%)   HAC z  raw p  Holm-adjus
ted p
        d_Fri      -22.3440 [−26.61%, −17.83%] −8.7647 0.0000
0.0000
start_of_month       −0.3145   [−3.42%, +2.89%] −0.1951 0.8453
0.8453
   last_5_days       −1.9662   [−5.60%, +1.81%] −1.0298 0.3031
0.6062
```

Under Holm adjustment for the three-test family, only the Friday effect rejects at α = 0.05 (Holm p < 0.001). Neither month-phase effect is significant under Holm; the raw p-values are already above 0.05. The Friday × regime section below asks whether the −22.3% average conceals regime variation.

Payday reframe: ruling out a practically meaningful start-of-month effect

Testing the start-of-month coefficient against a null of β = 0 asks whether *any* effect exists. That is not the interesting question here — for staffing and stock planning, a small effect is invisible; what matters is whether a large-enough effect is present to warrant action. Take a +10% start-of-month lift as the smallest effect that would matter in practice, and test the one-sided null

H_0: β_start-of-month ≥ log(1.10) ≈ 0.0953

against H_1: β_start-of-month < log(1.10). Rejecting H_0 means the data formally rule out a practically meaningful positive start-of-month effect. The end-of-month coefficient can be read the same way — its 95% one-sided lower bound gives the largest end-of-month dip the data are consistent with.

In [8]:
```
BETA_MIN_EFFECT = float(np.log(1.10))  # +10% on the original revenue sc
ale

# Start-of-month one-sided test
beta_hat_start = float(model.params["start_of_month"])
se_start = float(model.bse["start_of_month"])
z_start = (beta_hat_start - BETA_MIN_EFFECT) / se_start
p_start = float(stats.norm.cdf(z_start))
upper_start = beta_hat_start + 1.6449 * se_start

# End-of-month (last-5-days): show the largest dip the data allow
beta_hat_end = float(model.params["last_5_days"])
se_end = float(model.bse["last_5_days"])
lower_end = beta_hat_end - 1.6449 * se_end  # 95% one-sided lower

print("Start-of-month, one-sided test against β ≥ log(1.10):")
print(
    f"  β̂                                   : {beta_hat_start:+.4f}
({(np.exp(beta_hat_start) - 1) * 100:+.2f}%)"
)
print(f"  HAC SE                                : {se_start:.4f}")
print(f"  z                                     : {z_start:+.2f}")
print(f"  one-sided p                           : {p_start:.3g}")
print(
    f"  95% one-sided upper CI              : {upper_start:+.4f}  ({(np.
exp(upper_start) - 1) * 100:+.2f}%)"
)
print()
if upper_start < BETA_MIN_EFFECT:
    print("  The 95% upper CI is below the +10% threshold: reject H₀ at
α = 0.05.")
    print(
```

```
        "  The data rule out a +10% or larger start-of-month effect on c
ounter cash receipts."
    )

print()
print("Last-five-days, one-sided lower CI on the effect size:")
print(
    f"  β̂                                   : {beta_hat_end:+.4f}  ({(n
p.exp(beta_hat_end) - 1) * 100:+.2f}%)"
)
print(
    f"  95% one-sided lower CI              : {lower_end:+.4f}  ({(np.ex
p(lower_end) - 1) * 100:+.2f}%)"
)
print(
    f"  → the data are consistent with an end-of-month dip up to about "
    f"{abs((np.exp(lower_end) - 1) * 100):.1f}% but no larger."
)
```

```
Start-of-month, one-sided test against β ≥ log(1.10):
  β̂                                   : -0.0032  (-0.31%)
  HAC SE                              : 0.0162
  z                                   : -6.10
  one-sided p                         : 5.43e-10
  95% one-sided upper CI              : +0.0234  (+2.37%)

  The 95% upper CI is below the +10% threshold: reject H₀ at α = 0.05.
  The data rule out a +10% or larger start-of-month effect on counter cash
receipts.

Last-five-days, one-sided lower CI on the effect size:
  β̂                                   : -0.0199  (-1.97%)
  95% one-sided lower CI              : -0.0516  (-5.03%)
  → the data are consistent with an end-of-month dip up to about 5.0% but
no larger.
```

Friday-effect stability across regimes

NB02 removed weekly seasonality by subtracting a single day-of-week mean before searching for structural breaks. That step assumes the day-of-week pattern is stable across regimes. This section refits the joint calendar model with Friday × regime interactions to check that assumption and runs a Wald test on the joint null that all three interaction coefficients are zero. The regime-specific Friday effects (β_Friday + β_Friday×regime) are reported alongside.

In [9]:
```
df["fri_x_acute"] = df["d_Fri"] * df["r_acute"]
df["fri_x_rebound"] = df["d_Fri"] * df["r_rebound"]
df["fri_x_plateau"] = df["d_Fri"] * df["r_plateau"]

REGRESSORS_INT = REGRESSORS + ["fri_x_acute", "fri_x_rebound", "fri_x_pl
ateau"]
X_int = sm.add_constant(df[REGRESSORS_INT])
model_int = sm.OLS(y, X_int).fit(cov_type="HAC", cov_kwds={"maxlags": 2
1})

wald = model_int.wald_test(
    "fri_x_acute = fri_x_rebound = fri_x_plateau = 0", scalar=True, use_
```

```
f=False
)
print("Wald test for Friday × regime interactions all zero (HAC 21):")
print(f"  χ²(3) = {wald.statistic:.2f}, p = {wald.pvalue:.3g}")

week_groups = pd.factorize(df.index.to_period("W-SAT"))[0]  # Sunday-Saturday weeks, the local working week
print()
print("Friday × regime, covariance-choice robustness:")
print(f"  {'covariance':<22} χ²(3)   p-value")
for label, kw in [
    ("classical", {}),
    ("HAC lag 7", {"cov_type": "HAC", "cov_kwds": {"maxlags": 7}}),
    ("HAC lag 21", {"cov_type": "HAC", "cov_kwds": {"maxlags": 21}}),
    ("clustered by week", {"cov_type": "cluster", "cov_kwds": {"groups": week_groups}}),
]:
    m = sm.OLS(y, X_int).fit(**kw)
    w = m.wald_test(
        "fri_x_acute = fri_x_rebound = fri_x_plateau = 0", scalar=True, use_f=False
    )
    print(f"  {label:<22} {w.statistic:>6.2f}  {w.pvalue:.4g}")

base_fri = model_int.params["d_Fri"]
regime_fri = {
    "Pre-COVID": base_fri,
    "Acute crash": base_fri + model_int.params["fri_x_acute"],
    "Partial rebound": base_fri + model_int.params["fri_x_rebound"],
    "Lower plateau": base_fri + model_int.params["fri_x_plateau"],
}
regime_fri_df = pd.DataFrame(
    {
        "regime": list(regime_fri.keys()),
        "Friday effect (log)": list(regime_fri.values()),
        "exp(β)−1 (%)": [(np.exp(b) - 1) * 100 for b in regime_fri.values()],
    }
)
print()
with pd.option_context(
    "display.float_format", "{:+.4f}".format, "display.max_colwidth", None
):
    print("Friday effect by regime (interaction model, vs Tuesday):")
    print(regime_fri_df.to_string(index=False))

lb_main = sm.stats.acorr_ljungbox(model.resid, lags=[7], return_df=True).iloc[0][
    "lb_stat"
]
lb_int_val = sm.stats.acorr_ljungbox(model_int.resid, lags=[7], return_df=True).iloc[0][
    "lb_stat"
]
print()
print(
    f"Ljung-Box(7): main model = {lb_main:.1f}, interaction model = {lb_int_val:.1f}."
)
```

```python
print()
print(
    "Primary-family and holiday coefficient stability (main model vs int
eraction model):"
)
for name in ["start_of_month", "last_5_days", "ramadan", "eid_al_adha"]:
    print(
        f"  {name:>18}: main β = {model.params[name]:+.4f}, "
        f"interaction β = {model_int.params[name]:+.4f}, "
        f"Δ = {model_int.params[name] - model.params[name]:+.4f}"
    )
```

```
Wald test for Friday × regime interactions all zero (HAC 21):
  χ²(3) = 66.18, p = 2.81e-14

Friday × regime, covariance-choice robustness:
  covariance             χ²(3)   p-value
  classical               17.57  0.0005384
  HAC lag 7               31.17  7.834e-07
  HAC lag 21              66.18  2.805e-14
  clustered by week       29.66  1.626e-06

Friday effect by regime (interaction model, vs Tuesday):
         regime  Friday effect (log)  exp(β)-1 (%)
      Pre-COVID              -0.1315      -12.3211
    Acute crash              -0.3614      -30.3321
Partial rebound              -0.2963      -25.6436
  Lower plateau              -0.2349      -20.9364

Ljung-Box(7): main model = 100.6, interaction model = 113.0.

Primary-family and holiday coefficient stability (main model vs interactio
n model):
      start_of_month: main β = -0.0032, interaction β = -0.0016, Δ = +0.00
16
         last_5_days: main β = -0.0199, interaction β = -0.0209, Δ = -0.00
10
             ramadan: main β = -0.0325, interaction β = -0.0310, Δ = +0.00
16
         eid_al_adha: main β = -0.5119, interaction β = -0.4962, Δ = +0.01
57
```

The Wald test rejects the joint null of no Friday × regime interactions strongly ($\chi^2(3) \approx 66$ under HAC 21, $p < 10^{-13}$), and it rejects under every other covariance choice tested: classical $p = 0.0005$, HAC 7 $p < 10^{-6}$, cluster-by-week $p < 10^{-5}$. Friday depression was not stable across regimes: pre-COVID Fridays were about −12% versus Tuesday, acute-crash Fridays about −30%, partial-rebound Fridays about −26%, lower-plateau Fridays about −21% (all relative to Tuesday, from the Friday-only interaction model).

The primary-family and religious-calendar coefficients barely move between the main model and the interaction model, and the interaction-model Ljung-Box(7) is somewhat higher than the main model's (113 against 101): adding Friday × regime does not further reduce residual autocorrelation. The remaining autocorrelation is unlikely to be a weekday pattern the model has yet to capture; NB03's detrending

check shows within-regime drift is present, which would also produce the residual autocorrelation seen here.

The next cell rebuilds NB02's break-date search under a regime-aware day-of-week correction to check whether NB02's break dates change under a regime-aware correction.

In [10]:
```
# NB02 break-date check under a regime-aware day-of-week correction.
# Regime-aware: subtract the within-regime day-of-week deviation only, l
eaving
# regime levels intact so the break search sees the same level structur
e.
from pharmacy_covid_analysis.bootstrap import fit_ssr

idx_local = df.index
regime_idx = pd.Series(
    sum(
        (idx_local >= pd.Timestamp(b)).astype(int)
        for b in ["2020-03-24", "2020-05-28", "2020-12-04"]
    ),
    index=idx_local,
)
log_y_series = df["log_y"]
pooled_dow = log_y_series.groupby(idx_local.dayofweek).transform("mean")
cell_mean = log_y_series.groupby([regime_idx, idx_local.dayofweek]).tran
sform("mean")
regime_mean = log_y_series.groupby(regime_idx).transform("mean")

signal_nb02_naive = (log_y_series - pooled_dow).values
signal_nb02_aware = (log_y_series - (cell_mean - regime_mean)).values

_, bkps_naive = fit_ssr(signal_nb02_naive, 3)
_, bkps_aware = fit_ssr(signal_nb02_aware, 3)
print("NB02 K=3 break dates:")
print(f"  pooled-dow (current): {[log_y_series.index[i].date() for i in
bkps_naive]}")
print(f"  regime-aware:         {[log_y_series.index[i].date() for i in
bkps_aware]}")
```

```
NB02 K=3 break dates:
  pooled-dow (current): [datetime.date(2020, 3, 24), datetime.date(2020,
5, 28), datetime.date(2020, 12, 4)]
  regime-aware:         [datetime.date(2020, 3, 24), datetime.date(2020,
5, 28), datetime.date(2020, 12, 4)]
```

What the check shows. K=3 break dates are identical under both corrections (24 March, 28 May, 4 December). Because the missing weeks remove one of every weekday, each regime holds nearly equal numbers of each weekday, so the pooled and the regime-aware corrections give the same regime *means* (they differ only by a constant), so the break search sees the same signal. NB02's break structure does not depend on the pooled day-of-week correction. The material implication of the Friday × regime finding is for NB03's residual variance ratios; NB03 has a dedicated regime-aware robustness section that reruns the primary tests under the same correction.

Falsification test: whether only Friday changed across regimes

From running the pharmacy, I can suggest a specific mechanism for the Friday depression: Friday is the first day of the local weekend and doctors' surgeries are closed, so Friday sales are mostly non-prescription over-the-counter products, the most discretionary category and the one the COVID-19 restrictions hit hardest. About 20% of Fridays ran on reduced hours throughout the sample, so operating hours are not the driver of the *change* across regimes.

If this mechanism is right, the deepening should be **specific to Friday**: the other days should not show regime-varying interactions of comparable size, because their prescription share is much higher and prescription demand is not discretionary. The test adds a day × regime interaction for every day except Friday (and Tuesday, which is the base) and tests jointly that all fifteen interactions are zero. The test is reported under several covariance choices, since Wald tests with HAC standard errors and 15 restrictions can behave unstably at this sample size.

In [11]:
```
OTHER_DAYS = ["Mon", "Wed", "Thu", "Sat", "Sun"]  # Tuesday is the base
for d in OTHER_DAYS:
    for r in ["acute", "rebound", "plateau"]:
        df[f"{d}_x_{r}"] = df[f"d_{d}"] * df[f"r_{r}"]
OTHER_INTERACTIONS = [
    f"{d}_x_{r}" for d in OTHER_DAYS for r in ["acute", "rebound", "plat
eau"]
]

X_full = sm.add_constant(df[REGRESSORS_INT + OTHER_INTERACTIONS])
week_groups = pd.factorize(df.index.to_period("W-SAT"))[0]  # Sunday-Sat
urday weeks, the local working week

print("Joint Wald test on the 15 non-Friday day-by-regime interaction
s:")
print(f"  {'covariance':<22} χ²(15)  p-value")
for label, kw in [
    ("classical", {}),
    ("HAC lag 7", {"cov_type": "HAC", "cov_kwds": {"maxlags": 7}}),
    ("HAC lag 14", {"cov_type": "HAC", "cov_kwds": {"maxlags": 14}}),
    ("HAC lag 21", {"cov_type": "HAC", "cov_kwds": {"maxlags": 21}}),
    ("HAC lag 28", {"cov_type": "HAC", "cov_kwds": {"maxlags": 28}}),
    ("clustered by week", {"cov_type": "cluster", "cov_kwds": {"groups":
week_groups}}),
]:
    m = sm.OLS(y, X_full).fit(**kw)
    w = m.wald_test(" = ".join(OTHER_INTERACTIONS) + " = 0", scalar=Tru
e, use_f=False)
    print(f"  {label:<22} {w.statistic:>6.2f}  {w.pvalue:.4g}")

# Base-independent day effects: each day's coefficient relative to the n
on-Friday mean
# within its regime. Uses the HAC 21 fit for the point estimates.
model_full = sm.OLS(y, X_full).fit(cov_type="HAC", cov_kwds={"maxlags":
21})
day_effects = {}
```

```
for d in ["Mon", "Wed", "Thu", "Fri", "Sat", "Sun"]:
    row = {"Pre-COVID": model_full.params[f"d_{d}"]}
    for r in ["acute", "rebound", "plateau"]:
        key = f"fri_x_{r}" if d == "Fri" else f"{d}_x_{r}"
        row[r] = model_full.params[f"d_{d}"] + model_full.params[key]
    day_effects[d] = row
day_effects["Tue"] = {"Pre-COVID": 0.0, "acute": 0.0, "rebound": 0.0, "p
lateau": 0.0}
eff_df = pd.DataFrame(day_effects).T
non_fri_mean = eff_df.drop(index="Fri").mean(axis=0)
rel = eff_df.sub(non_fri_mean, axis=1).apply(lambda col: (np.exp(col) -
1) * 100)
rel.columns = ["Pre-COVID", "Acute crash", "Partial rebound", "Lower pla
teau"]
print()
print("Day-of-week effect relative to the non-Friday mean within each re
gime (%):")
print("(from the full interaction model, HAC 21)")
with pd.option_context("display.float_format", "{:+.2f}".format):
    print(rel.reindex(["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Su
n"]).to_string())
```

```
Joint Wald test on the 15 non-Friday day-by-regime interactions:
  covariance             χ²(15)  p-value
  classical               10.57  0.7823
  HAC lag 7               16.82  0.3297
  HAC lag 14              25.61  0.04234
  HAC lag 21              41.80  0.0002416
  HAC lag 28              48.42  2.179e-05
  clustered by week       20.37  0.1583

Day-of-week effect relative to the non-Friday mean within each regime (%):
(from the full interaction model, HAC 21)
     Pre-COVID  Acute crash  Partial rebound  Lower plateau
Mon      -1.42        +0.39            -2.45          +1.04
Tue      -1.18        +5.30            +0.85          -5.65
Wed      +1.99        -5.35            -3.18          +0.59
Thu      -1.26        -4.97            +0.43          -0.71
Fri     -13.16       -30.92           -26.30         -21.59
Sat      +0.59        +4.45            +2.17          +2.10
Sun      +1.33        +0.69            +2.31          +2.86
```

Reading the joint test. Friday × regime rejects under every covariance choice, from classical ($p = 0.0005$) to cluster-by-week ($p < 10^{-5}$) to both HAC bandwidths tested (7 and 21). The 15-parameter non-Friday joint test behaves very differently: it does not reject under classical ($p = 0.78$), short-lag HAC (7 lags, $p = 0.33$), or cluster-by-week ($p = 0.16$); the χ^2 rises steadily with HAC bandwidth (17, 26, 42, 48 at lags 7, 14, 21, 28), unlike the primary coefficients, whose HAC standard errors rise with bandwidth (diagnostics below). That rising pattern suggests an unstable HAC covariance estimate for these 15 sparse interaction terms. Even classical standard errors, which ignore autocorrelation and tend to over-reject, find nothing ($p = 0.78$); short-lag HAC and cluster-by-week also fail to reject. Under the notebook's primary covariance, HAC 21, the test does reject ($p = 0.0002$); I set that result aside on the grounds above, which is a judgement about the estimator's stability rather than a diagnostic, and a block bootstrap of the Wald statistic would be the principled check. So the joint

pattern of non-Friday changes is not evident under any covariance choice that is stable here, but it is not ruled out either.

What the base-independent table shows. Under the non-Friday mean reference, Friday moves about 18 percentage points across regimes (about −13% before COVID to about −31% in the acute crash). Between pre-COVID and the acute crash the largest other-day move is Wednesday, about 7 percentage points (about +2% to about −5%); across all four regimes Tuesday's range is wider, about 11 points (about +5% in the crash to about −6% on the plateau; Tuesday is the base of the Friday-only model, so its base-independent value is minus the mean of the other five days' interactions). None shows the consistent, sustained shift Friday does. So Friday's change is robust to every SE choice and larger than any other day's movement; the other days' change shows up only with HAC at 14 lags or more, where the covariance estimate looks unstable, so it is unproven rather than ruled out.

Diagnostics

Compact diagnostic panel on the joint-model residuals: Q-Q plot, studentised residuals over time, Breusch–Pagan homoskedasticity test, HAC lag sensitivity for the primary Friday and month-phase coefficients.

In [12]:
```
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

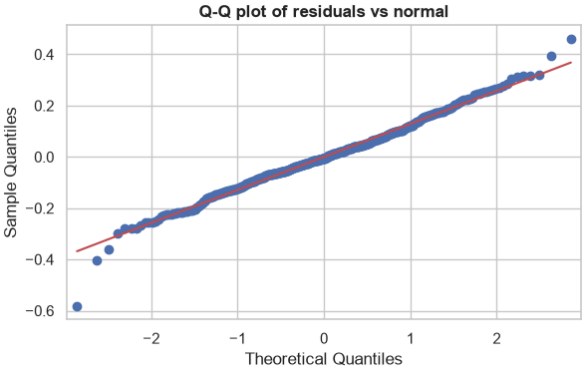
# Q-Q plot
sm.qqplot(model.resid, line="s", ax=axes[0])
axes[0].set_title("Q-Q plot of residuals vs normal")

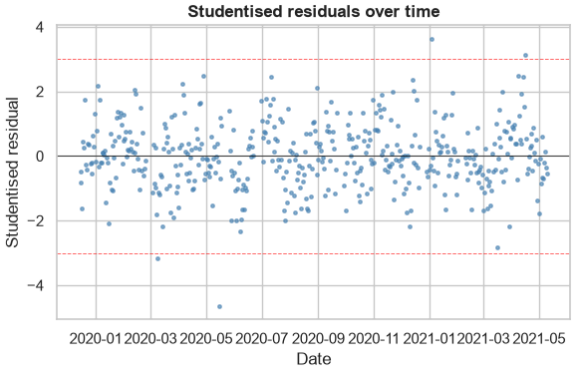
# Studentised residuals over time
stud = model.outlier_test()["student_resid"]
axes[1].scatter(df.index, stud, s=6, alpha=0.6, color="steelblue")
axes[1].axhline(0, color="black", linewidth=0.5)
axes[1].axhline(3, color="red", linestyle="--", linewidth=0.7, alpha=0.
6)
axes[1].axhline(-3, color="red", linestyle="--", linewidth=0.7, alpha=0.
6)
axes[1].set_title("Studentised residuals over time")
axes[1].set_xlabel("Date")
axes[1].set_ylabel("Studentised residual")
plt.tight_layout()
plt.savefig("../charts/nb04_residual_diagnostics.png", dpi=150, bbox_inc
hes="tight")
plt.show()

# Dates with |studentised residual| > 3
extremes = stud[stud.abs() > 3].sort_values()
print("Dates with |studentised residual| > 3:")
for d, v in extremes.items():
    print(f"  {d.date()} ({d.day_name()}): {v:+.2f}")

# Breusch-Pagan
bp = sm.stats.het_breuschpagan(model.resid, model.model.exog)
print()
print(f"Breusch-Pagan LM = {bp[0]:.2f}, p = {bp[1]:.4f}")
```

```python
# HAC lag sensitivity for Friday and month-phase coefficients
lag_rows = []
for L in [7, 14, 21, 28]:
    m = sm.OLS(y, X).fit(cov_type="HAC", cov_kwds={"maxlags": L})
    for name in ["d_Fri", "start_of_month", "last_5_days"]:
        lag_rows.append(
            {
                "coefficient": name,
                "HAC lag": L,
                "β": m.params[name],
                "HAC SE": m.bse[name],
                "HAC z": m.tvalues[name],
                "p-value": m.pvalues[name],
            }
        )
lag_df = pd.DataFrame(lag_rows)
with pd.option_context("display.float_format", "{:.4f}".format):
    print()
    print("HAC lag sensitivity for the primary-family coefficients:")
    print(
        lag_df.pivot(
            index="coefficient", columns="HAC lag", values=["β", "HAC SE", "p-value"]
        ).to_string()
    )
```






```
Dates with |studentised residual| > 3:
  2020-05-15 (Friday): −4.65
  2020-03-09 (Monday): −3.16
  2021-04-16 (Friday): +3.12
  2021-01-03 (Sunday): +3.63

Breusch–Pagan LM = 11.18, p = 0.5957

HAC lag sensitivity for the primary-family coefficients:
                    β                          HAC SE
p-value
HAC lag             7       14      21      28     7      14     21     28
7      14     21     28
coefficient
d_Fri          −0.2529 −0.2529 −0.2529 −0.2529 0.0251 0.0270 0.0289 0.0309
0.0000 0.0000 0.0000 0.0000
last_5_days    −0.0199 −0.0199 −0.0199 −0.0199 0.0188 0.0190 0.0193 0.0193
0.2900 0.2956 0.3031 0.3028
start_of_month −0.0032 −0.0032 −0.0032 −0.0032 0.0186 0.0168 0.0162 0.0158
0.8654 0.8515 0.8453 0.8419
```

The Q-Q plot shows heavier-than-normal tails, driven by four days with |studentised residual| > 3: 2020-05-15 (Friday during the acute crash, mid-Ramadan; the largest deviation at −4.7; this is also NB01's single lowest day in the sample), 2021-01-03 (Sunday, lower plateau; the first working day after the New Year weekend, and New Year is not modelled here), 2021-04-16 (Friday, lower plateau, Ramadan) and 2020-03-09 (Monday, pre-COVID; 15 days before the acute-crash break, and lining up with NB01's finding that revenue over 3–23 March 2020 was already about 10% below the January–February average). Breusch–Pagan does not reject homoskedasticity of the residuals at conventional levels. HAC lag sensitivity is small: the Friday and month-phase HAC standard errors and p-values are stable across bandwidths from 7 to 28 lags (the OLS coefficients do not depend on the bandwidth).

Reconciliation with NB01's descriptive figures

NB01 reported descriptive comparisons on the raw revenue scale, without controlling for regime, month-phase or the religious calendar:

NB01 figure	NB01 value	NB04 estimate under the joint model
Friday vs non-Friday mean	−22.7% (−16.6% after removing the lowest 20% of Fridays)	Friday vs Tuesday: −22.3% (average across regimes)
Start-of-month vs mid-month	−1.4%	Start-of-month vs mid-month: −0.3% (not significant)
Last five days vs mid-month	−5.2%	Last five days vs mid-month: −2.0% (not significant)

The Friday averages line up closely because Monday, Wednesday and Thursday are near-zero relative to Tuesday, so "vs Tuesday" and "vs non-Friday mean" produce almost the same number here. But the average conceals substantial regime variation: pre-COVID Fridays were only about −13% under the base-independent framing (see the falsification table); Friday's shift dominates all other days across regimes. NB01's Friday explanation (first day of the local weekend, a demand pattern) is consistent with the *presence* of a pre-COVID Friday dip but does not by itself explain the deepening after March 2020. NB01 cross-references this finding.

The month-phase figures shrink towards zero because NB01's raw comparisons did not control for regime levels or the religious calendar; adding those controls absorbs most of the raw month-phase differences.

Summary

All findings describe counter cash receipts (about 65% of revenue; see NB01), analysed on log revenue with 25 December 2019 excluded. The joint model fits regime + day-of-week + month-phase + religious-calendar dummies in one specification with HAC (Newey–West) standard errors at a 21-lag bandwidth. Compared with a single COVID dummy, the four-regime structure cuts the Ljung-Box

statistic at lag 7 from about 690 to about 100; residual autocorrelation remains positive but the regime dummies did most of the correction.

Primary family (Holm-adjusted, exploratory). Three tests chosen after the NB01 EDA. Friday effect (−22.3% average vs Tuesday, HAC z = −8.8, Holm $p < 0.001$) rejects; start-of-month (−0.3%, Holm $p \approx 0.85$) and last-five-days (−2.0%, Holm $p \approx 0.61$) do not.

Payday reframe. Taking a +10% start-of-month lift as the smallest effect that would matter in practice, the one-sided test (H_0: $\beta \geq \log(1.10)$) gives $p < 10^{-9}$ and a 95% one-sided upper CI of +2.4% — well below +10%. The data rule out a practically meaningful positive start-of-month effect on counter cash receipts. The end-of-month coefficient similarly rules out a dip larger than about 5.0%.

Friday changed across regimes; other days did not clearly change. Friday × regime rejects under every covariance choice ($\chi^2 > 17$ with $p < 0.001$ under classical SEs, HAC 7, cluster-by-week and HAC 21). The 15-parameter non-Friday joint test does not reject with short-lag HAC ($p = 0.33$) or standard errors clustered by week ($p = 0.16$), and even classical standard errors, which ignore autocorrelation and tend to over-reject, find nothing ($p = 0.78$). It rejects only with HAC at 14 lags or more, where the χ^2 rises steadily with the bandwidth, which suggests an unstable HAC covariance for these sparse interaction terms; since HAC 21 is the primary covariance elsewhere in this notebook, setting that rejection aside is a judgement, and the other days' change is unproven rather than ruled out. Base-independent day effects show Friday moving about 18 points (from about −13% pre-COVID to about −31% in the acute crash), against about 7 for Wednesday between pre-COVID and the crash and about 11 for Tuesday across all four regimes, neither of them sustained. Friday sales are mostly non-prescription over-the-counter products (from my own knowledge of the business); about 20% of Fridays ran reduced hours consistently across the sample, so the *change* is not operational. The series has no prescription/non-prescription split, so the mechanism cannot be tested directly.

Implication for NB02. Repeating NB02's break search with a regime-aware day-of-week correction leaves the K=3 break dates unchanged (24 March, 28 May, 4 December): the regime means are the same under both corrections because the missing weeks remove one of every weekday, leaving nearly equal weekday counts in each regime. NB02's break structure does not depend on the pooled day-of-week correction. The material implication is for NB03's variance ratios; NB03 has a dedicated regime-aware robustness section that reruns its primary tests under the same correction.

Religious calendar and NB01 reconciliation. Ramadan is small and not significant (−3.2%, HAC z = −1.29). Eid al-Adha 2020 is estimated at −40% relative to a Friday–Saturday weekend the model already controls for; the coefficient picks up NB01's noted reduced trading hours on 31 July and the extra Friday depression in the rebound regime not captured by the pooled Friday effect, identified by only two observations. Other local public holidays (New Year, Labour Day, etc.) are not

modelled. The Friday figure matches NB01 closely on average (–22.3% here vs NB01's descriptive –22.7%); the base-independent breakdown adds that the dip roughly doubled after March 2020, which NB01's Friday paragraph cross-references.